Chapter 9: Rational Functions

MULTIPLE CHOICE

1. Which of the reciprocal functions has a vertical asymptote with equation $x = \frac{9}{2}$?

A. $f(x) = \frac{1}{2x + 9}$

B. $f(x) = \frac{1}{2x - 9}$

C. $f(x) = \frac{9}{x + 2}$

D. $f(x) = \frac{1}{x + \frac{9}{2}}$

ANS: B DIF: Easy OBJ: Section 9.1 NAT: RF14
TOP: Exploring Rational Functions Using Transformations
KEY: reciprocal of linear function | vertical asymptote

2. What is true about the behaviour of the function $f(x) = \frac{1}{4x + 5}$ as $x \to -\frac{5}{4}^{+}$ (right to left)?

A. $f(x) \to -\infty$

B. $f(x) \to +\infty$

C. $f(x) \to 0$

D. $f(x)$ is undefined

ANS: B DIF: Average OBJ: Section 9.1 NAT: RF14
TOP: Exploring Rational Functions Using Transformations
KEY: reciprocal of linear function | behaviour at non-permissible values

3. What is the x-intercept of $f(x) = \frac{1}{2x + 4}$?

A. There is no x-intercept.

B. $-\frac{1}{2}$

C. –2

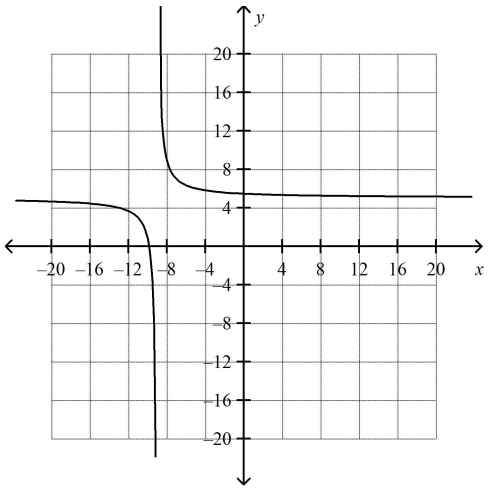
D. 0

ANS: A DIF: Easy OBJ: Section 9.1 NAT: RF14
TOP: Exploring Rational Functions Using Transformations
KEY: reciprocal of linear function | x-intercept

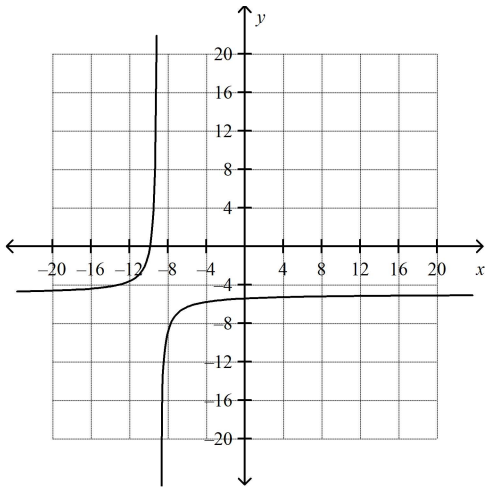
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4. Which graph represents the function $f(x) = \dfrac{-4}{x-9} - 5$?

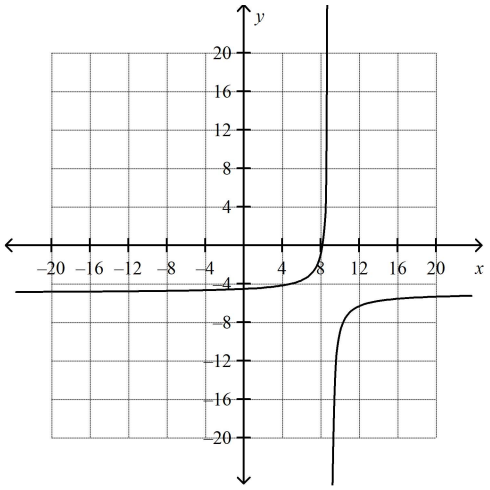
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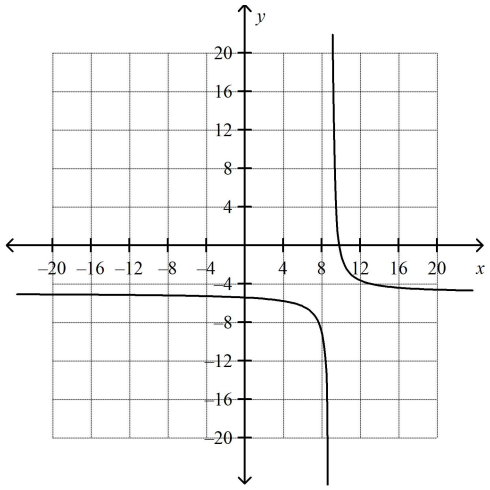
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B.



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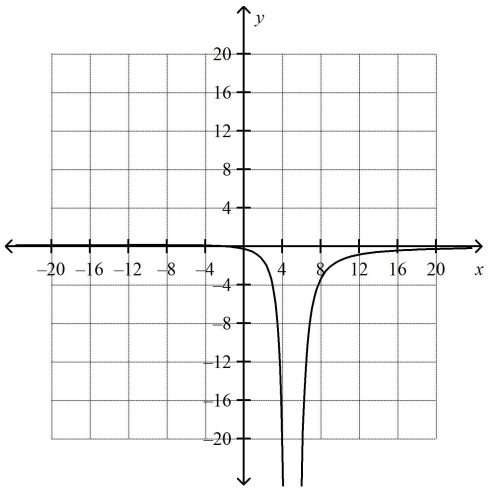


ANS: B DIF: Average OBJ: Section 9.1 NAT: RF14
TOP: Exploring Rational Functions Using Transformations
KEY: reciprocal of linear function | graph from function

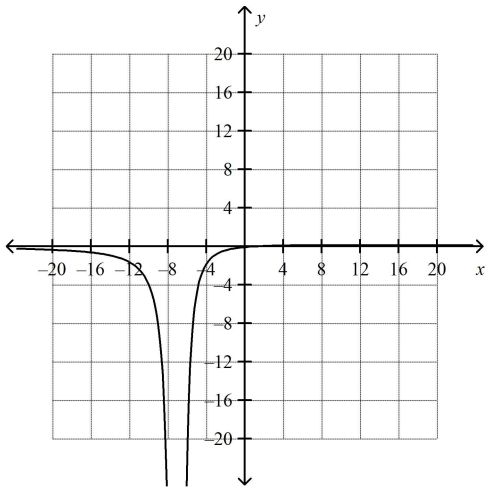
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5. The rational function $f(x) = \dfrac{-3x-7}{(x-5)^2}$ is represented by which graph?

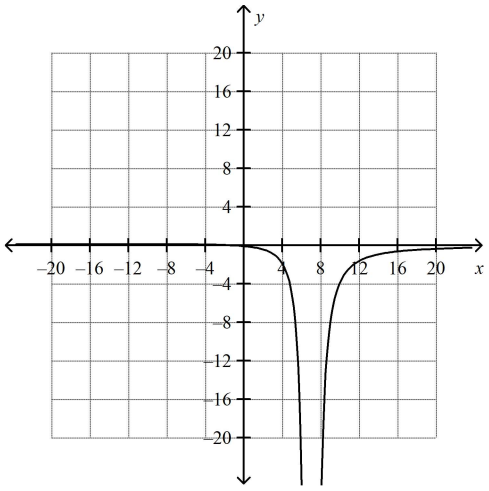
A.



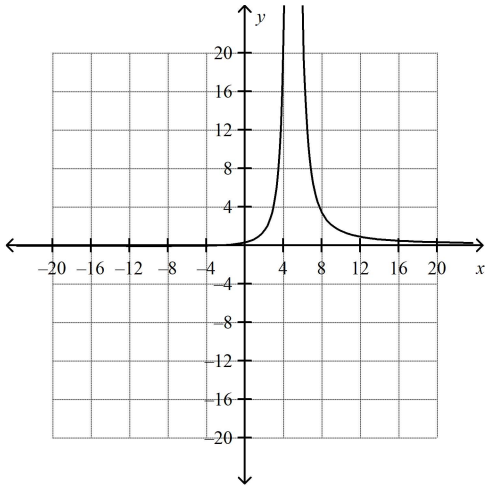
C.



B.



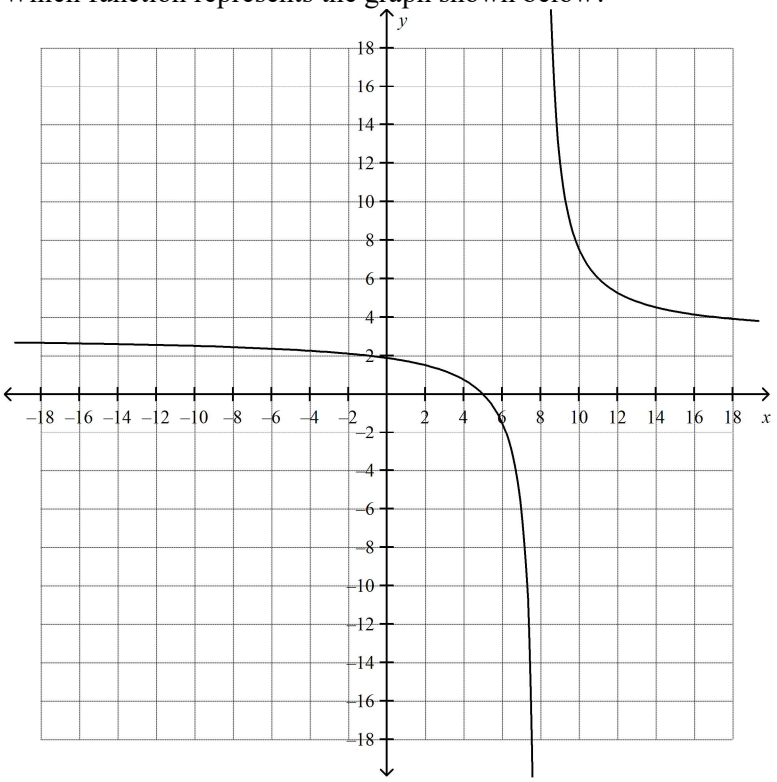
D.



ANS: A DIF: Difficult OBJ: Section 9.1 NAT: RF14
TOP: Exploring Rational Functions Using Transformations
KEY: reciprocal of linear function | graph from function

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6. Which function represents the graph shown below?



A. $f(x) = \dfrac{9}{x-8} + 3$

B. $f(x) = \dfrac{-9}{x-8} + 3$

C. $f(x) = \dfrac{-9}{x+3} - 8$

D. $f(x) = \dfrac{9}{x+3} - 8$

ANS: A DIF: Average OBJ: Section 9.1 NAT: RF14
TOP: Exploring Rational Functions Using Transformations
KEY: reciprocal of linear function | function from graph

7. Which of the following functions has a slant asymptote when graphed?

A. $f(x) = \dfrac{5x^3 - 10x^2 - 15x}{x^2 - 3x - 4}$

B. $f(x) = \dfrac{5x^3 - 10x^2 - 15x}{x^2 - 2x - 3}$

C. $f(x) = \dfrac{5x^3 - 10x^2 - 15x}{x^2 - 3x}$

D. all of the above

ANS: A DIF: Difficult + OBJ: Section 9.1 NAT: RF14
TOP: Exploring Rational Functions Using Transformations KEY: slant asymptote | hole | factor

8. What is the value of k in the function $f(x) = \dfrac{2-k}{5x+k}$ if its graph passes through the point $(3, -\dfrac{2}{19})$?

A. $-\dfrac{19}{2}$

B. 4

C. –10

D. No such k exists

ANS: B DIF: Difficult OBJ: Section 9.1 NAT: RF14
TOP: Exploring Rational Functions Using Transformations KEY: reciprocal of linear function

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9. Which function has vertical asymptotes with equations $x = -9$ and $x = -\frac{6}{7}$?

A. $f(x) = \frac{7x + 6}{x^2 + 15x + 54}$

B. $f(x) = \frac{1}{7x^2 + 69x + 54}$

C. $f(x) = \frac{1}{7x^2 - 69x + 54}$

D. $f(x) = \frac{-9}{x^2 + 15x + 54}$

ANS: B DIF: Average OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions
KEY: reciprocal of quadratic function | vertical asymptote

10. Which function has a point of discontinuity at $x = 3$?

A. $f(x) = \frac{x - 3}{2x^2 - 2x - 12}$

B. $f(x) = \frac{x + 3}{x^2 - 6x - 12}$

C. $f(x) = \frac{x - 3}{x^2 - 6x - 12}$

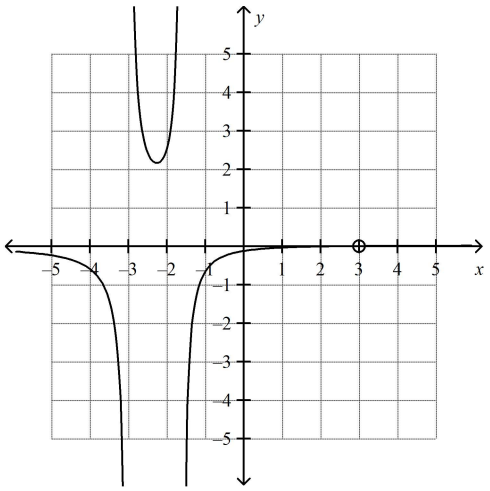
D. $f(x) = \frac{x + 3}{x^2 - 6x + 9}$

ANS: A DIF: Average OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions
KEY: reciprocal of quadratic function | vertical asymptote

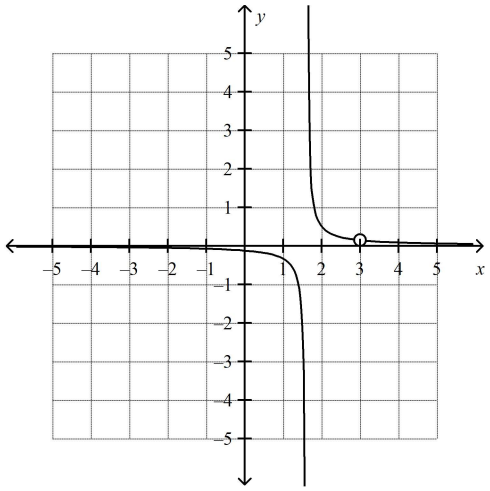
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11. Which graph represents $f(x) = \dfrac{x-3}{5x^2-23x+24}$?

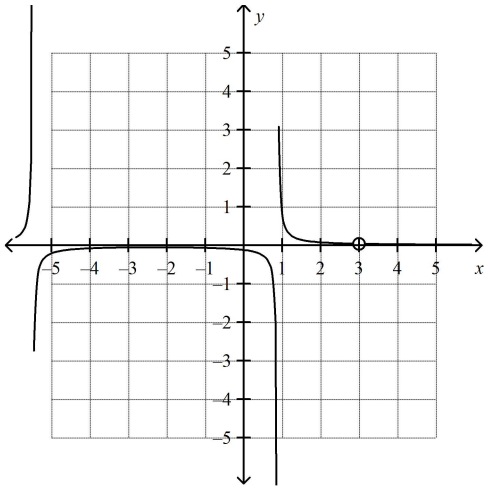
A.



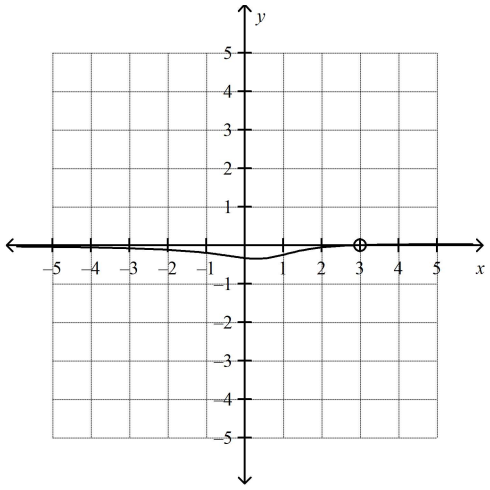
C.



B.



D.



ANS: C DIF: Difficult OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions KEY: rational function | discontinuity | hole

12. Which function has a y-intercept of $-\dfrac{8}{27}$?

A. $f(x) = \dfrac{-8}{x^2-12x-27}$

B. $f(x) = \dfrac{8}{(-8x+3)(x+9)}$

C. $f(x) = \dfrac{-8}{x^2+12x+27}$

D. all of the above

ANS: C DIF: Average OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions KEY: reciprocal of quadratic function | y-intercept

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13. Which function has a horizontal asymptote with equation $y = \frac{2}{7}$?

A. $f(x) = \frac{-2x-3}{7x+8}$

B. $f(x) = \frac{7x+8}{2x-3}$

C. $f(x) = \frac{7x-3}{2x+8}$

D. $f(x) = \frac{2x-3}{7x+8}$

ANS: D DIF: Average OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions
KEY: linear expressions in numerator and denominator | horizontal asymptote

14. Which function has an x-intercept of $\frac{1}{3}$?

A. $f(x) = \frac{-6x-2}{5x-3}$

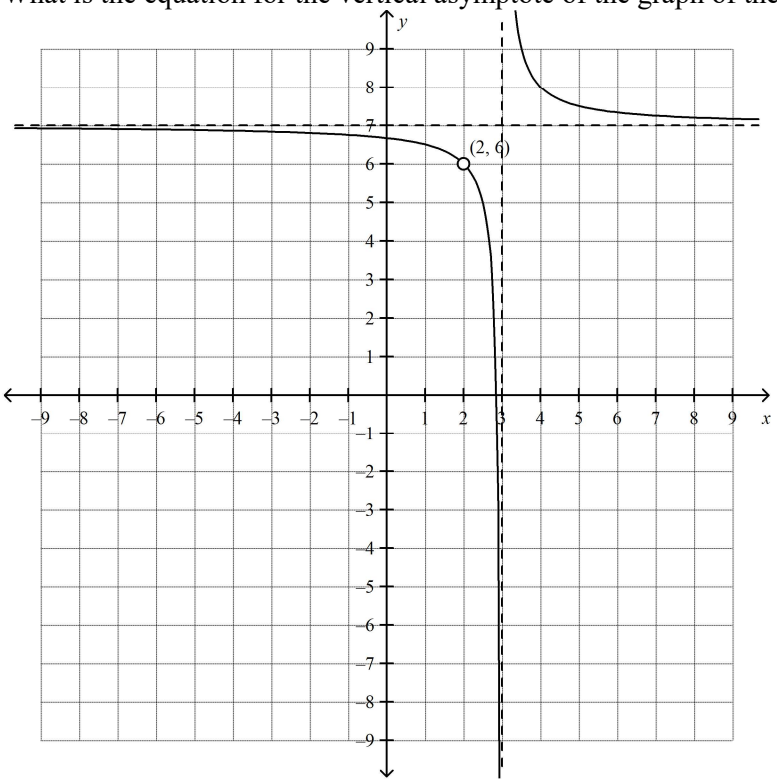
B. $f(x) = \frac{5x-3}{6x-2}$

C. $f(x) = \frac{6x-2}{5x-3}$

D. $f(x) = \frac{5x-2}{6x-3}$

ANS: C DIF: Average OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions
KEY: linear expressions in numerator and denominator | x-intercept

15. What is the equation for the vertical asymptote of the graph of the function shown?



A. $x = 2$

B. $x = 3$

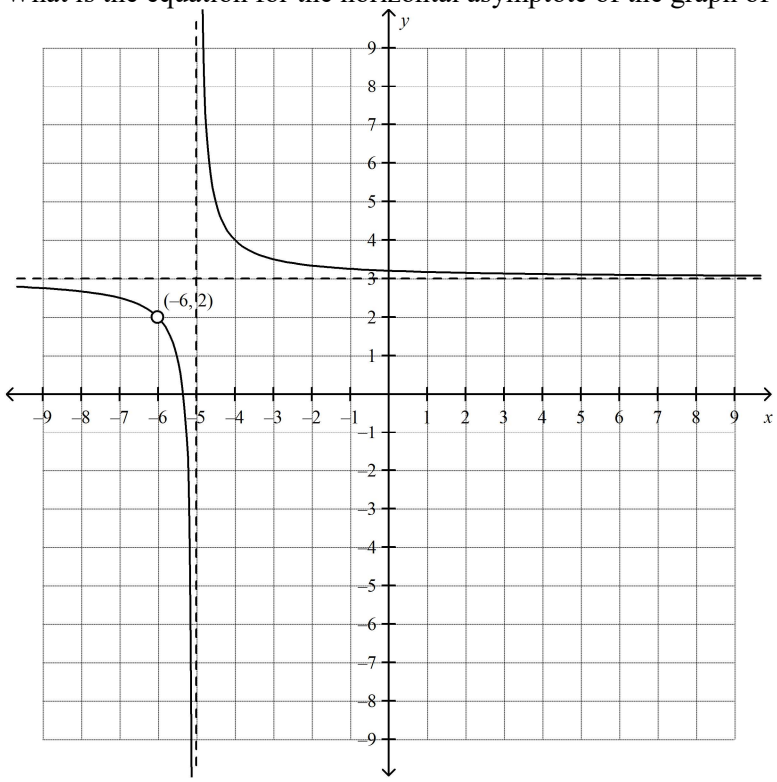
C. $y = 7$

D. $y = 6$

ANS: B DIF: Easy OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions KEY: quadratic denominator | vertical asymptote

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16. What is the equation for the horizontal asymptote of the graph of the function shown?



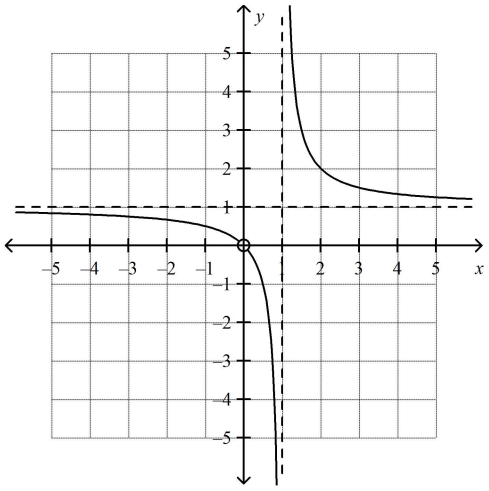
A. $y = 2$
B. $x = -5$
C. $y = 3$
D. $x = -6$

ANS: C DIF: Easy OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions KEY: quadratic denominator | horizontal asymptote

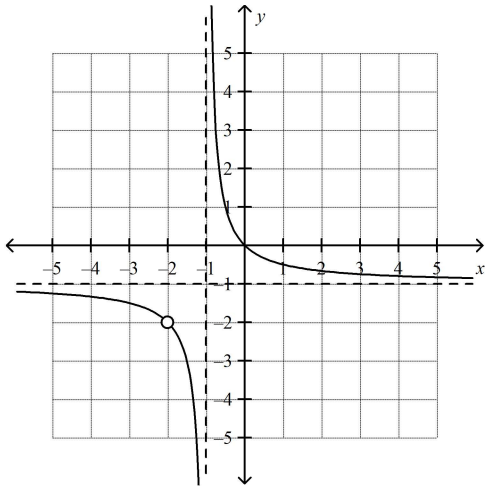
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17. Which graph of a rational function has the following characteristics?
- a vertical asymptote with equation $x = -1$
- a horizontal asymptote with equation $y = -1$
- a point of discontinuity at $(-2, -2)$

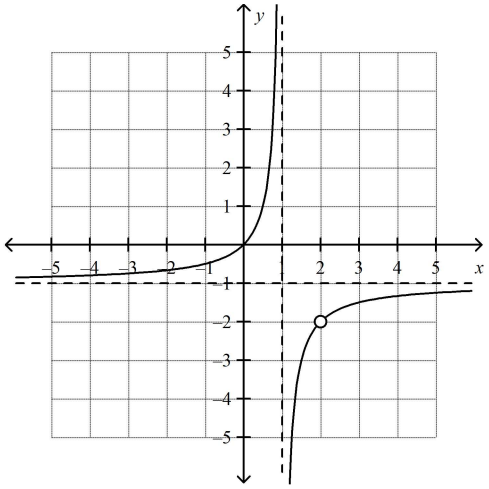
A.



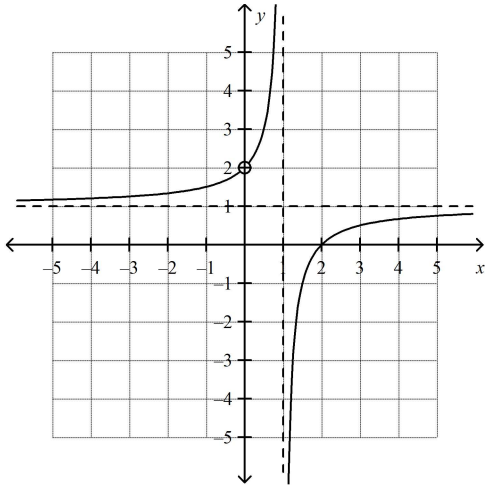
C.



B.



D.



ANS: C DIF: Average OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions KEY: quadratic denominator | graph from characteristics

18. Which rational function has the following characteristics?
- a vertical asymptote with equation $x = 2$
- a horizontal asymptote with equation $y = 8$
- a point of discontinuity at $(3, 9)$

A. $f(x) = -\dfrac{x-3}{(x+3)(x-2)} - 8$

C. $f(x) = \dfrac{x+3}{(x+2)(x+3)} + 8$

B. $f(x) = \dfrac{x-3}{(x-3)(x-2)} + 8$

D. $f(x) = \dfrac{x-2}{(x-3)(x+2)} + 8$

ANS: B DIF: Difficult OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions
KEY: quadratic denominator | function from characteristics

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19. Which function has a graph in the shape of a parabola?

A. $f(x) = \dfrac{(x-3)^2(x-7)}{(x-3)(x-7)}$

B. $f(x) = \dfrac{(x-3)^2(x-7)}{x-3}$

C. $f(x) = \dfrac{x-3}{(x-3)^3(x-7)}$

D. none of the above

ANS: B DIF: Average OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions KEY: hole

20. Solve the equation $\dfrac{-1}{x+6} = 8+x$.

A. $x = -7$

B. $x = -8$

C. $x = -6$

D. $x = -\dfrac{1}{7}$

ANS: A DIF: Easy OBJ: Section 9.3 NAT: RF14
TOP: Connecting Graphs and Rational Equations KEY: rational equation

21. Solve the equation $\dfrac{-2}{x+6} = \dfrac{3}{x}$.

A. $x = \dfrac{18}{5}$

B. $x = 3$

C. $x = -\dfrac{18}{5}$

D. $x = 4$

ANS: C DIF: Average OBJ: Section 9.3 NAT: RF14
TOP: Connecting Graphs and Rational Equations KEY: rational equation

22. Solve the equation $\dfrac{x+3}{x-8} = \dfrac{x+8}{x+9}$.

A. $x = -3$

B. $x = \dfrac{91}{12}$

C. $x = -\dfrac{91}{12}$

D. $x = 8, x = -9$

ANS: C DIF: Average OBJ: Section 9.3 NAT: RF14
TOP: Connecting Graphs and Rational Equations KEY: rational equation

23. Solve the equation $\dfrac{x}{x+2} = \dfrac{9}{x+9}$.

A. $x = \dfrac{-9 \pm 2\sqrt{9}}{2}$

B. $x = 2, x = 9$

C. no solution

D. $x = \pm 3\sqrt{2}$

ANS: D DIF: Easy OBJ: Section 9.3 NAT: RF14
TOP: Connecting Graphs and Rational Equations KEY: rational equation

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24. What are the x-intercepts of the graph of $f(x) = \dfrac{x^2 + 7x - 18}{x^2 + 12x + 35}$?

A. –7, –5
B. 2, –9
C. 7, 5
D. –2, 9

ANS: B DIF: Average OBJ: Section 9.3 NAT: RF14
TOP: Connecting Graphs and Rational Equations KEY: rational function | x-intercept

25. Solve the equation $0 = \dfrac{6x^3 + 6}{-x^3 - 8x^2}$ graphically.

A. no solution
B. $x = -1$
C. 0
D. $x = -8$

ANS: B DIF: Difficult + OBJ: Section 9.3 NAT: RF14
TOP: Connecting Graphs and Rational Equations KEY: rational equation | graph

COMPLETION

1. The rational function $f(x) = \dfrac{1}{-17x - 17}$ has a vertical asymptote with equation _____.

ANS: $x = -1$

DIF: Easy OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions KEY: reciprocal of linear function | vertical asymptote

2. The rational function $f(x) = \dfrac{13}{x^2 - 11}$ has a y-intercept of _____.

ANS: $-\dfrac{13}{11}$

DIF: Easy OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions KEY: reciprocal of quadratic function | y-intercept

3. The number of x-intercepts in the graph of $f(x) = \dfrac{(x+3)(x+8)(x+4)}{x(x+8)}$ is _____.

ANS: 2

DIF: Easy OBJ: Section 9.3 NAT: RF14
TOP: Connecting Graphs and Rational Equations KEY: zeros | roots | x-intercept

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4. Given a rational function, if $x \neq a$ is a restriction, then $x = a$ could be the location of a point of discontinuity or the _____.

ANS: equation of a vertical asymptote

DIF: Average OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions KEY: point of discontinuity | vertical asymptote

5. When solving the rational function $\frac{x^2 - 8x - 6}{-4 + 3x} = x + 2$ graphically, the single function that could be graphed is _____.

ANS: $f(x) = \frac{x^2 - 8x - 6}{-4 + 3x} - x - 2$

DIF: Average OBJ: Section 9.3 NAT: RF14
TOP: Connecting Graphs and Rational Equations KEY: graph | rational function

MATCHING

Match each graph of a rational function with its equation.

A. $f(x) = \frac{9}{x^2 - 4}$

B. $f(x) = \frac{1}{x + 4}$

C. $f(x) = -\frac{1}{x + 4}$

D. $f(x) = \frac{1}{x^2 + 4}$

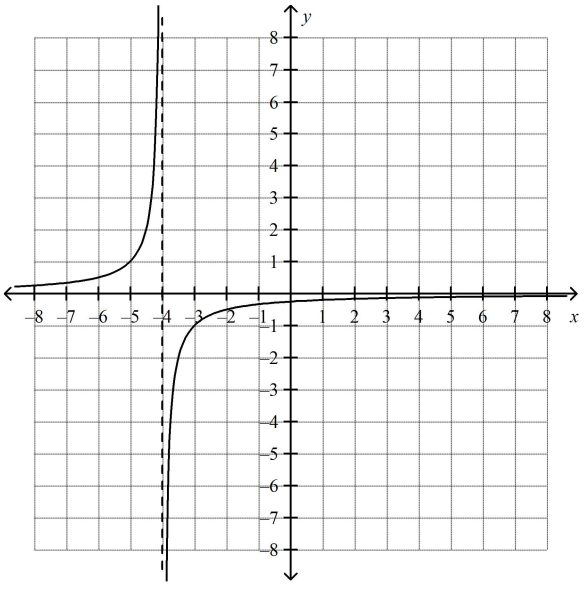
E. $f(x) = \frac{-3x - 9}{x + 4}$

F. $f(x) = \frac{9}{x^2 + 6x + 8}$

G. $f(x) = \frac{1}{(x + 4)^2}$

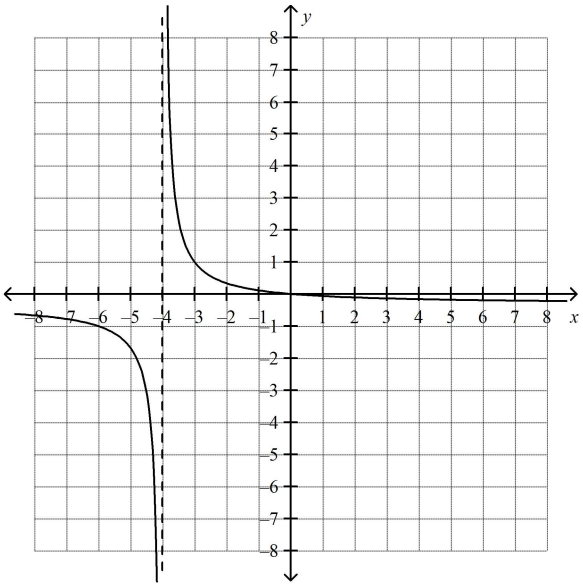
H. $f(x) = \frac{x}{-3(x + 4)}$

1.

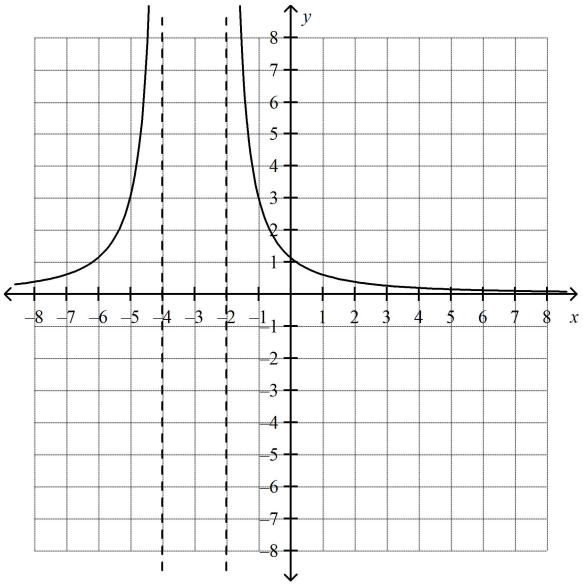


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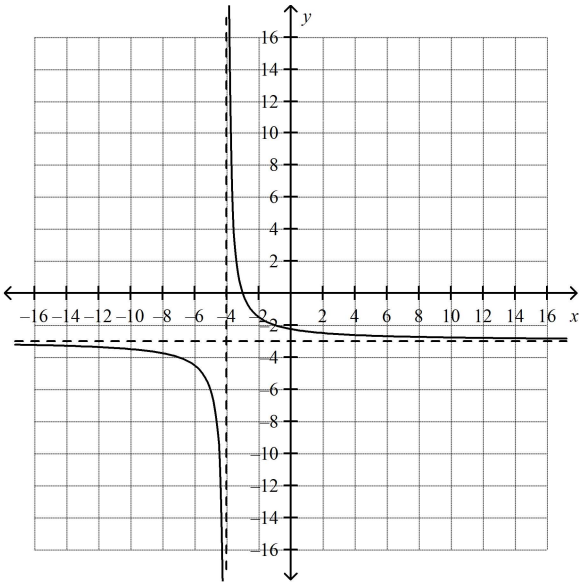
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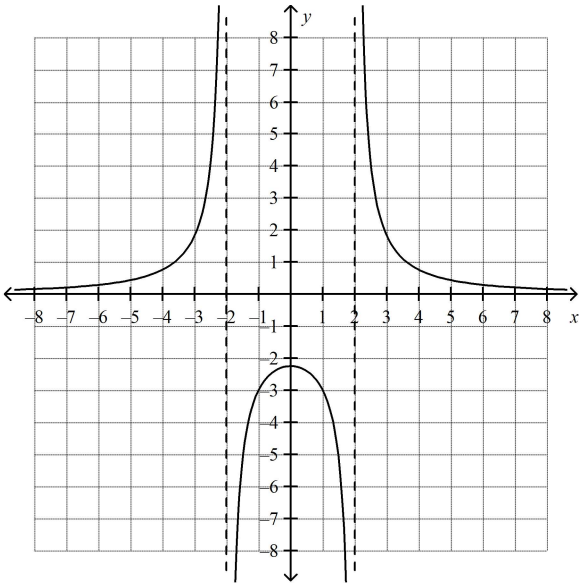


4.



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5.



1. ANS: C DIF: Average OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions KEY: rational functions | graphs

2. ANS: H DIF: Average OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions KEY: rational functions | graphs

3. ANS: F DIF: Average OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions KEY: rational functions | graphs

4. ANS: E DIF: Average OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions KEY: rational functions | graphs

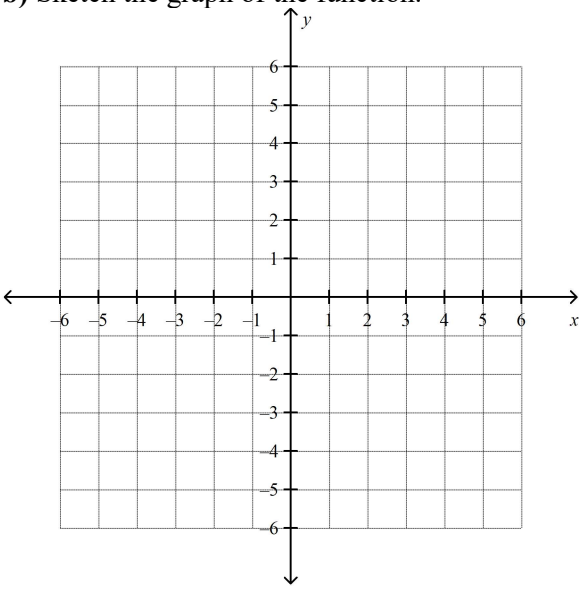
5. ANS: A DIF: Average OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions KEY: rational functions | graphs

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SHORT ANSWER

1. **a)** Determine an equation in the form $f(x) = \frac{1}{kx - c}$ for a function with a vertical asymptote at $x = 2$ and a y-intercept of $-\frac{1}{8}$.

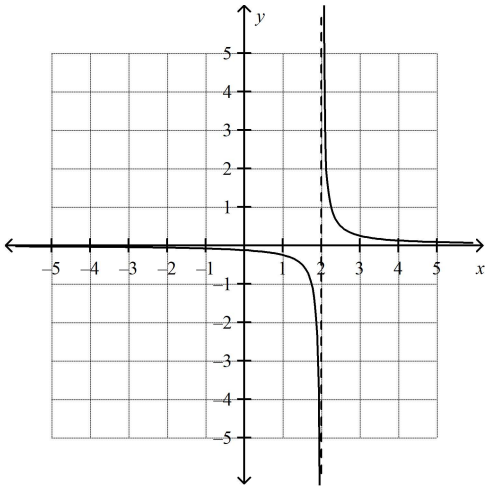
b) Sketch the graph of the function.



ANS:

a) $f(x) = \frac{1}{4x - 8}$

b)

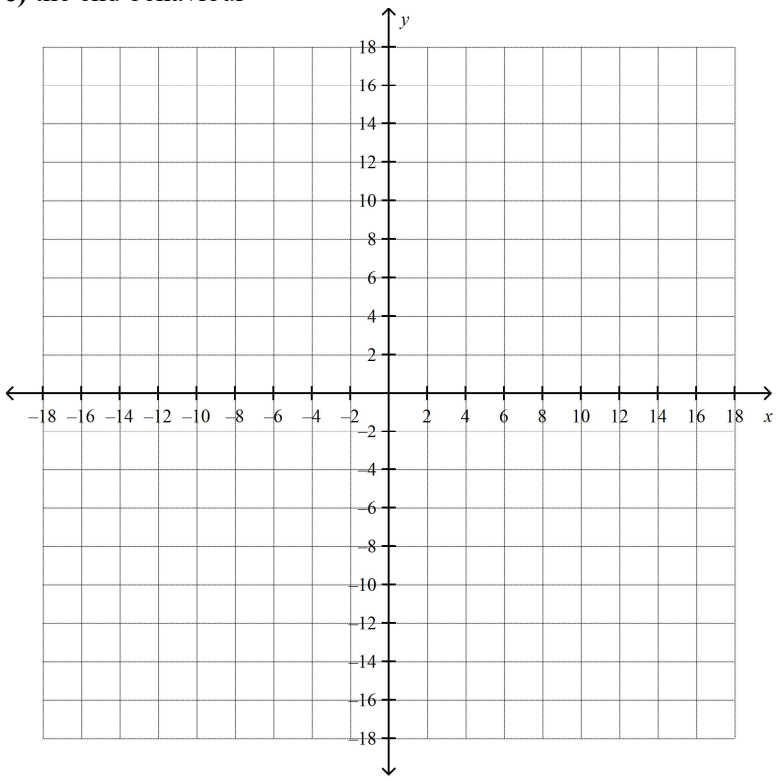


DIF: Average OBJ: Section 9.1 NAT: RF14
TOP: Exploring Rational Functions Using Transformations
KEY: reciprocal of linear function | vertical asymptote | y-intercept | graph

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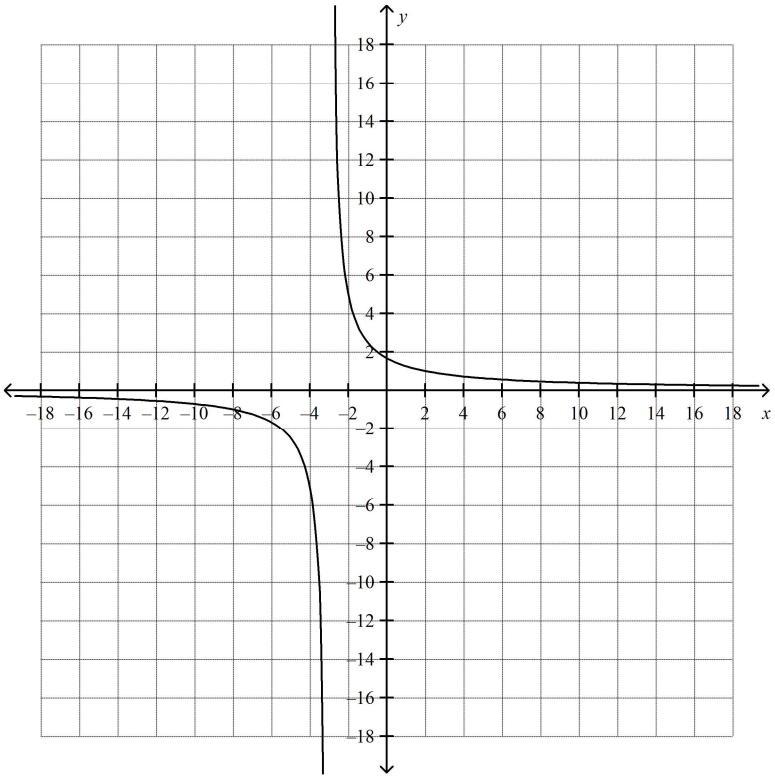
2. Sketch the graph of the function $f(x) = \dfrac{5}{x+3}$ using transformations and identify

a) the vertical asymptotes (if any)
b) the horizontal asymptotes (if any)
c) the domain and range
d) the behaviour near any non-permissible values
e) the end behaviour



ANS:

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a) vertical asymptote with equation $x = -3$

b) horizontal asymptote with equation $y = 0$

c) domain $\{x \mid x \neq -3, x \in \mathrm{R}\}$; range $\{y \mid y \neq -0, y \in \mathrm{R}\}$

d) As x approaches -3, $|y|$ becomes very large.

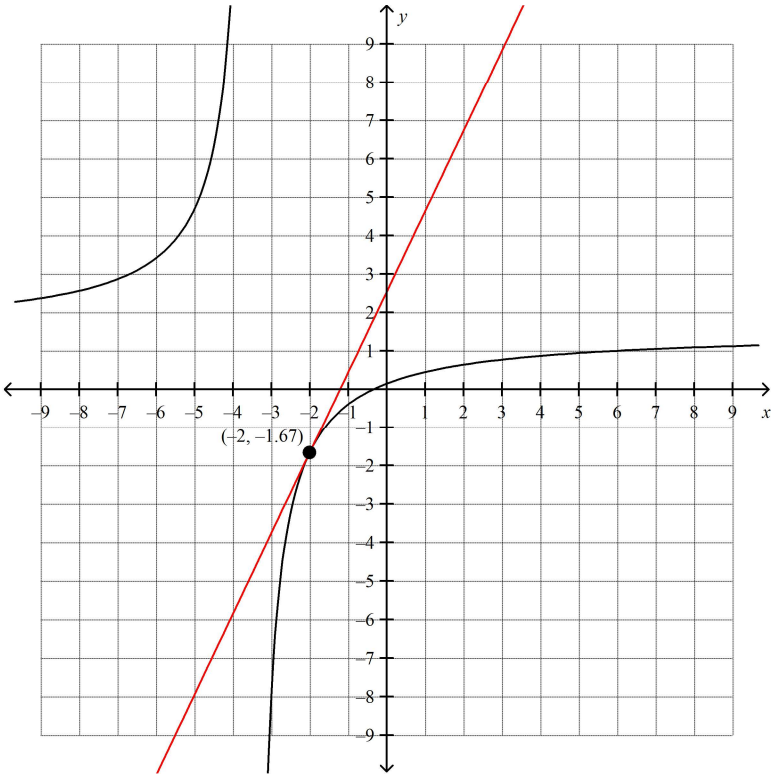
e) As $|x|$ becomes very large, y approaches 0.

DIF: Easy OBJ: Section 9.1 NAT: RF14
TOP: Exploring Rational Functions Using Transformations
KEY: graph | characteristics | transformations

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3. Determine the approximate slope of the tangent to the curve $f(x) = \dfrac{3x+1}{2x+7}$ at $x = -2$, to one decimal place.

ANS:
Use technology to draw the graph of the function and estimate the tangent line at the point of tangency, $\left(-2, -\dfrac{5}{3}\right)$.



$m \approx 2.1$

DIF: Difficult + OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions KEY: instantaneous rate of change

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4. Consider the function $f(x) = -\dfrac{2}{x+4}$.

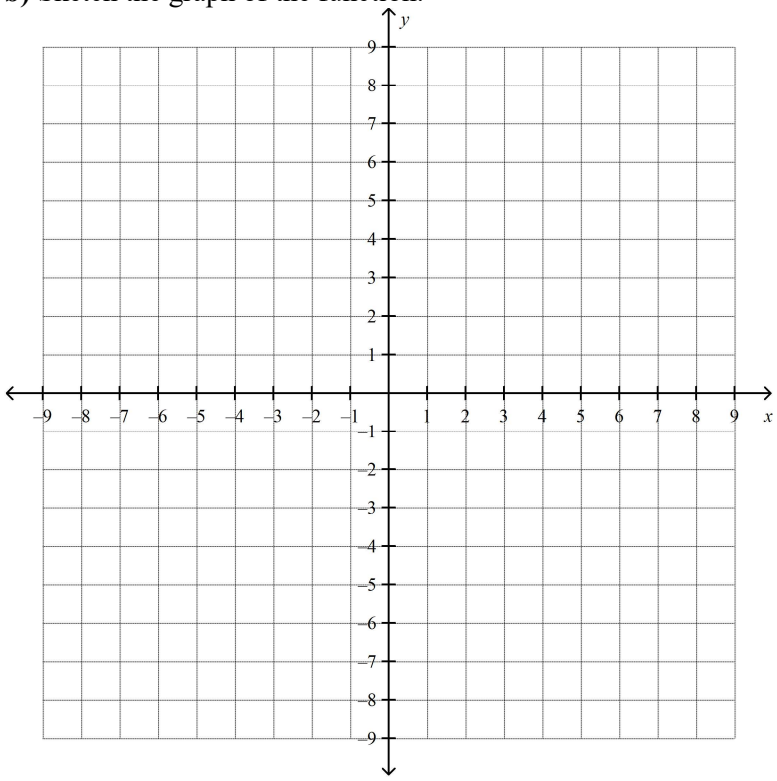
a) Determine the key features of the function:

i) domain and range

ii) intercepts

iii) equations of any asymptotes

b) Sketch the graph of the function.



ANS:

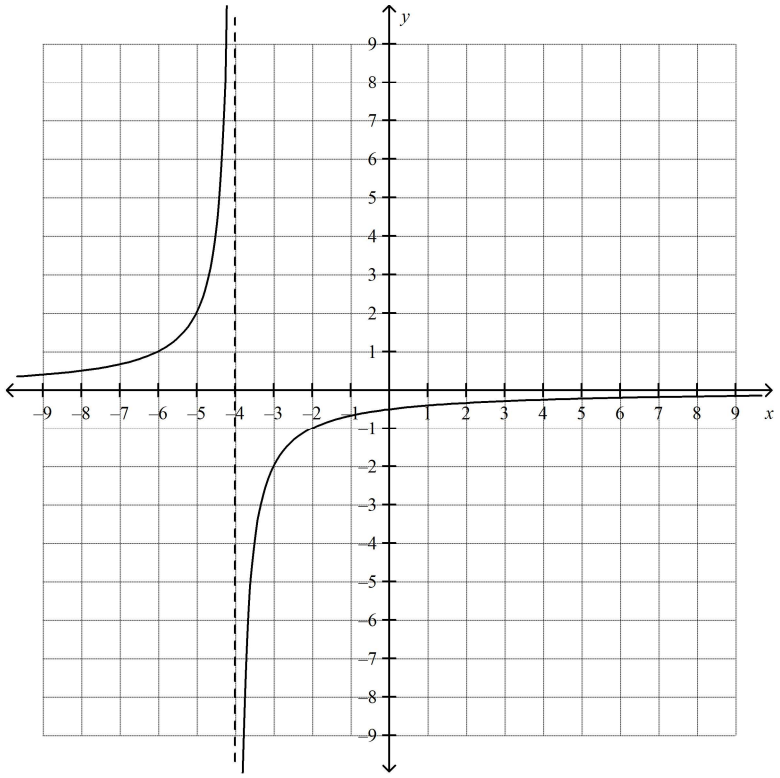
a) i) $\{x | x \neq -4, x \in \mathrm{R}\}$, $\{y | y \neq 0, y \in \mathrm{R}\}$

ii) x-intercept: none, y-intercept: $-\dfrac{1}{2}$

iii) $x = -4$, $y = 0$

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b)



DIF: Average OBJ: Section 9.2 | Section 9.3 NAT: RF14
TOP: Analysing Rational Functions | Connecting Graphs and Rational Equations
KEY: reciprocal of linear function | key features | graph

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5. Consider the function $f(x) = \dfrac{3}{4x-5}$.

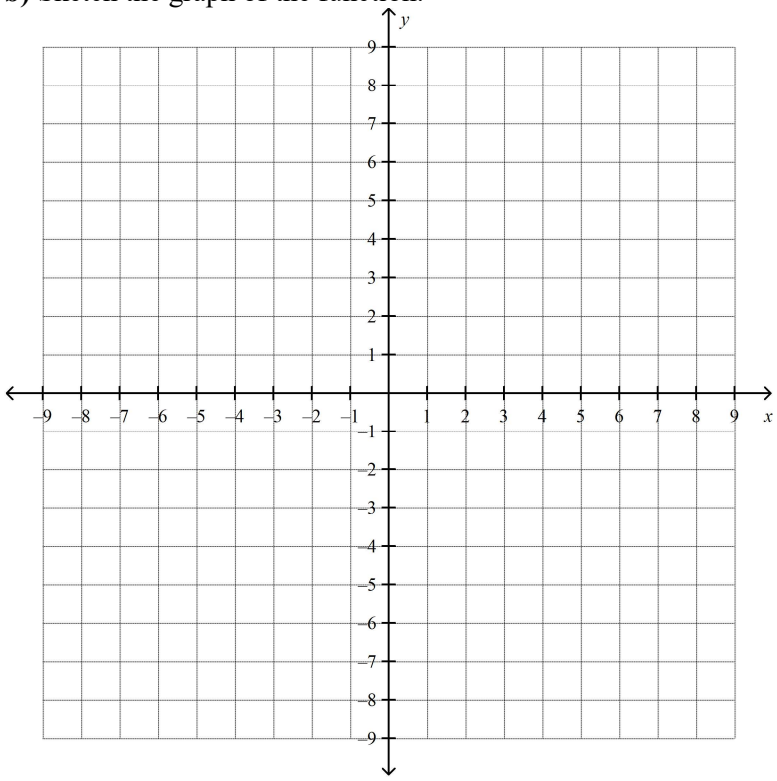
a) Determine the key features of the function:

i) domain and range

ii) intercepts

iii) equations of any asymptotes

b) Sketch the graph of the function.



ANS:

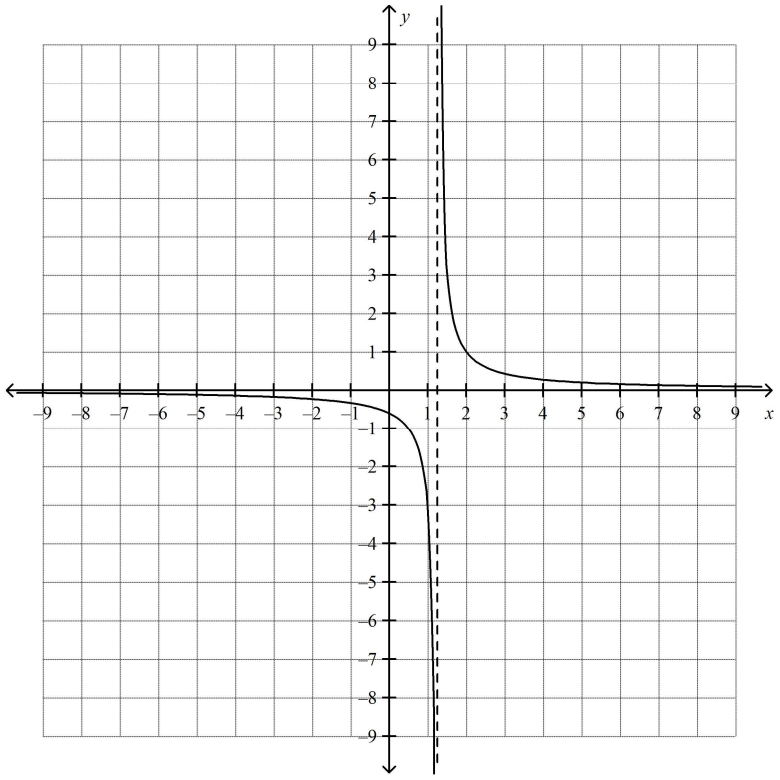
a) i) $\left\{x|x \neq \dfrac{5}{4}, x \in \mathrm{R}\right\}$, $\{y|y \neq 0, y \in \mathrm{R}\}$

ii) x-intercept: none, y-intercept: $-\dfrac{3}{5}$

iii) $x = \dfrac{5}{4}$, $y = 0$

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b)



DIF: Average OBJ: Section 9.2 | Section 9.3 NAT: RF14
TOP: Analysing Rational Functions | Connecting Graphs and Rational Equations
KEY: reciprocal of linear function | key features | graph

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6. Consider the function $f(x) = \dfrac{1}{x^2 + 2x - 8}$.

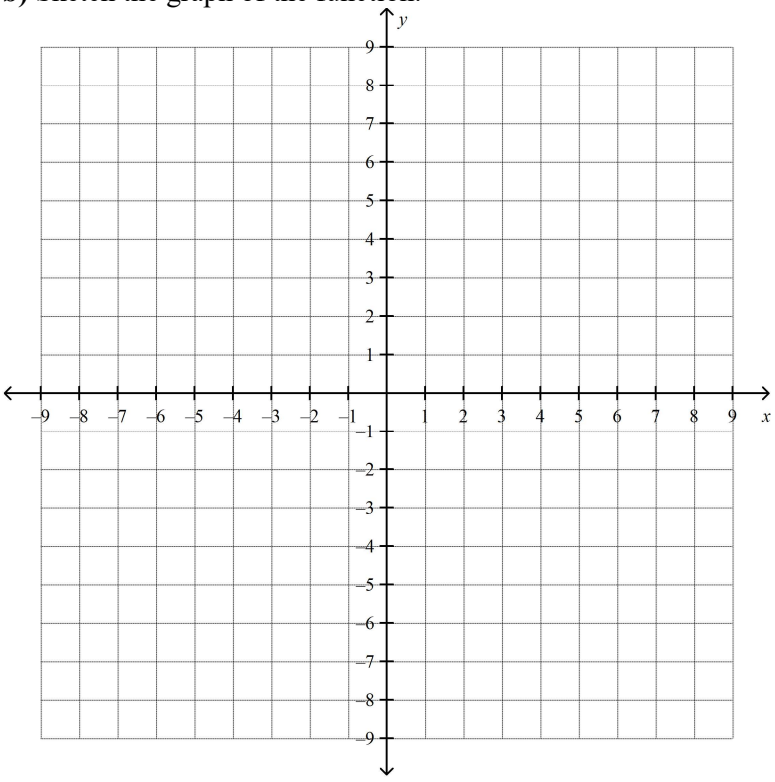
a) Determine the key features of the function:

i) domain and range

ii) intercepts

iii) equations of any asymptotes

b) Sketch the graph of the function.



ANS:

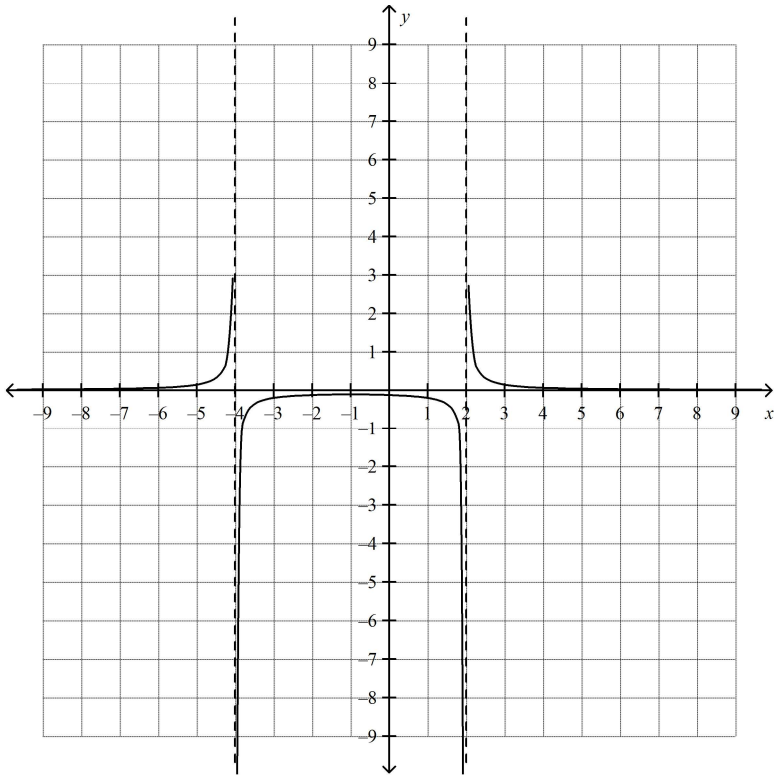
a) i) $\{x | x \neq 2, -4, x \in \mathrm{R}\}$, $\{y | y \neq 0, y \in \mathrm{R}\}$

ii) x-intercept: none, y-intercept: $-\dfrac{1}{8}$

iii) $x = 2, x = -4, y = 0$

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b)



DIF: Difficult OBJ: Section 9.2 | Section 9.3 NAT: RF14
TOP: Analysing Rational Functions | Connecting Graphs and Rational Equations
KEY: reciprocal of quadratic function | key features | graph

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7. Consider the function $f(x) = \dfrac{3x+8}{x-2}$.

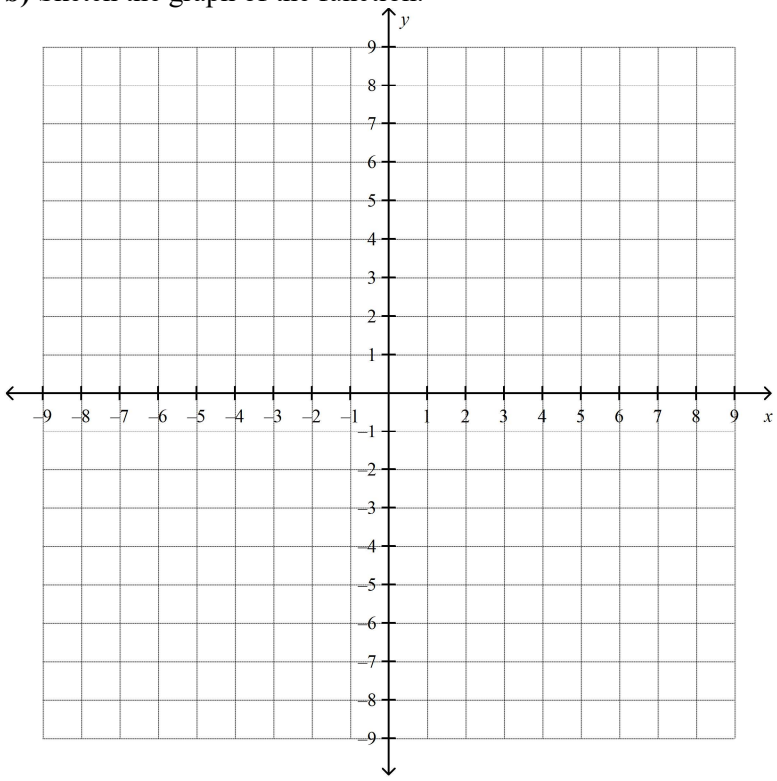
a) Determine the key features of the function:

i) domain and range

ii) intercepts

iii) equations of any asymptotes

b) Sketch the graph of the function.



ANS:

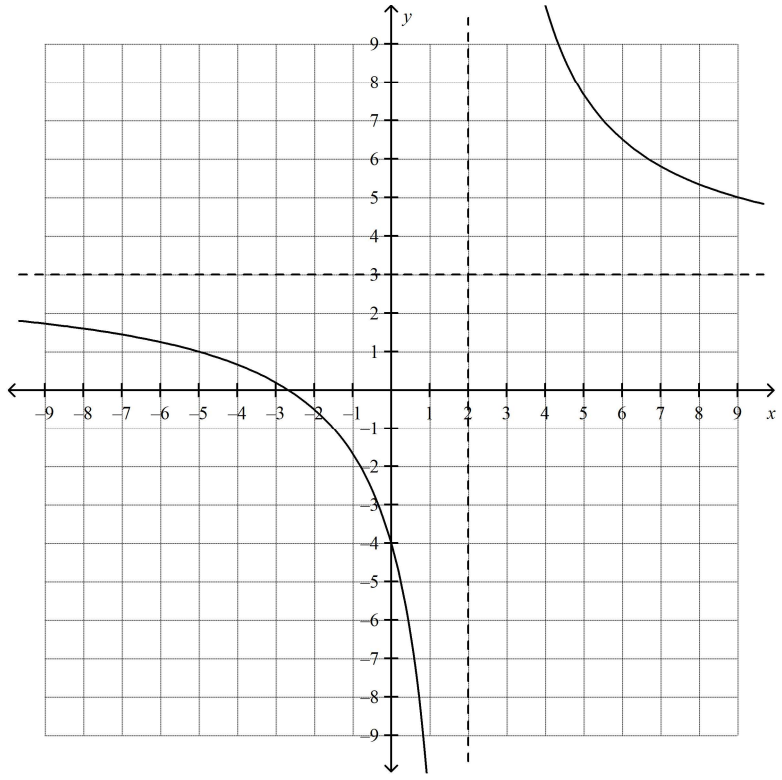
a) i) $\{x|x \neq 2, x \in \mathrm{R}\}$, $\{y|y \neq 3, y \in \mathrm{R}\}$

ii) x-intercept: $-\dfrac{8}{3}$, y-intercept: -4

iii) $x = 2, y = 3$

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b)



DIF: Average OBJ: Section 9.2 | Section 9.3 NAT: RF14
TOP: Analysing Rational Functions | Connecting Graphs and Rational Equations
KEY: linear expressions in numerator and denominator | key features | graph

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8. Consider the function $f(x) = \dfrac{x+3}{x^2 - x - 12}$.

a) Determine the key features of the function:
- **i)** domain and range
- **ii)** intercepts
- **iii)** equations of any asymptotes

b) Sketch the graph of the function.

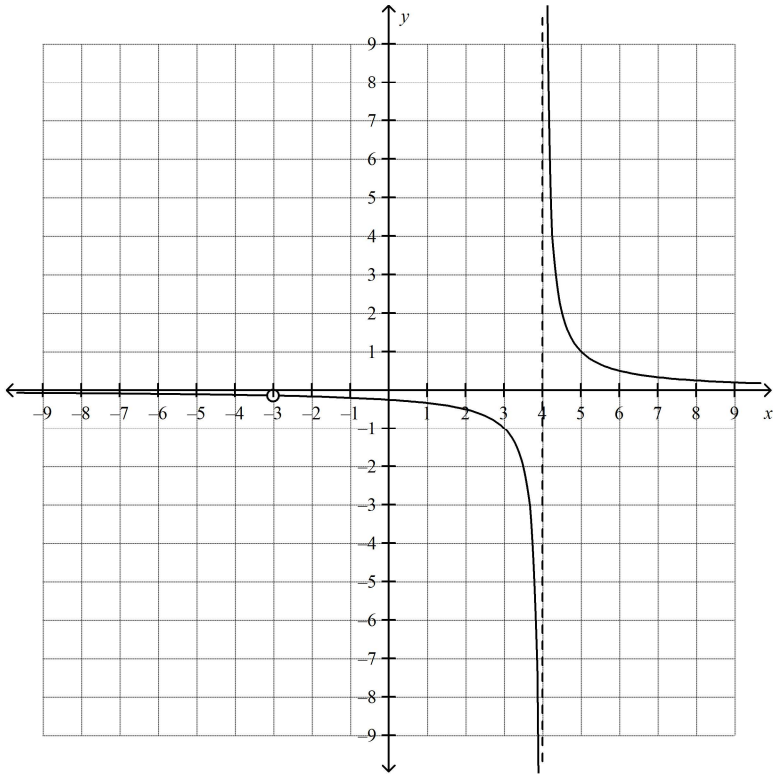
ANS:

a) i) $\{x | x \neq -3, x \neq 4, x \in \mathrm{R}\}$, $\{y | y \neq 0, y \in \mathrm{R}\}$

ii) x-intercept: none, y-intercept: $-\dfrac{1}{4}$

iii) $x = 4, y = 0$

b) Note the hole at the point $\left(-3, -\dfrac{1}{7}\right)$.



DIF: Difficult OBJ: Section 9.2 | Section 9.3 NAT: RF14
TOP: Analysing Rational Functions | Connecting Graphs and Rational Equations
KEY: rational function | key features | graph

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9. Solve.

a) $\frac{1}{x-4} = -\frac{6}{x}$

b) $\frac{x+5}{x-3} = \frac{2x+7}{x}$

c) $7 + \frac{1}{x} = \frac{1}{x-2}$

d) $\frac{3}{2x-4} = \frac{4}{x-2}$

ANS:

a) $\frac{1}{x-4} = -\frac{6}{x}$

$$x = -6(x-4)$$

$$x = -6x + 24$$

$$7x = 24$$

$$x = \frac{24}{7}$$

The solution satisfies the restrictions on x.

b) $\frac{x+5}{x-3} = \frac{2x+7}{x}$

$$x(x+5) = (x-3)(2x+7)$$

$$x^2 + 5x = 2x^2 + x - 21$$

$$x^2 - 4x - 21 = 0$$

$$(x-7)(x+3) = 0$$

$$x = 7 \text{ or } x = -3$$

Both solutions satisfy the restrictions on x.

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c) $7+\frac{1}{x}=\frac{1}{x-2}$

$$\frac{7x+1}{x}=\frac{1}{x-2}$$

$$(x-2)(7x+1)=x$$

$$7x^2-13x-2=x$$

$$7x^2-14x-2=0$$

$$x=\frac{-(-14)\pm\sqrt{(-14)^2-4(7)(-2)}}{2(7)}$$

$$=\frac{14\pm\sqrt{252}}{14}$$

$$=\frac{14\pm6\sqrt{7}}{14}$$

$$x=\frac{7\pm3\sqrt{7}}{7}$$

Both solutions satisfy the restrictions on x.

d) $\frac{3}{2x-4}=\frac{4}{x-2}$

$$3(x-2)=4(2x-4)$$

$$3x-6=8x-16$$

$$5x=10$$

$$x=2$$

But $x = 2$ is an inadmissible value for the equation, so there is no solution.

DIF: Average OBJ: Section 9.3 NAT: RF14
TOP: Connecting Graphs and Rational Equations KEY: rational equation

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10. Solve each equation by graphing, using technology. Express your answers to two decimal places.

a) $\frac{3x}{x-2} = -\frac{1}{x+3}$

b) $\frac{1}{x} + \frac{1}{x-5} = -1$

c) $\frac{x^2}{(x-5)^3} = -\frac{2}{x-5}$

d) $\frac{3x-1}{x} = \frac{x+5}{x-3}$

ANS:
a) $x \approx 0.19$ or $x \approx -3.52$
b) $x \approx -1.19$ or $x \approx 4.19$
c) No solution.
d) $x \approx 0.21$ or $x \approx 7.29$

DIF: Difficult OBJ: Section 9.3 NAT: RF14
TOP: Connecting Graphs and Rational Equations
KEY: rational equation | technology

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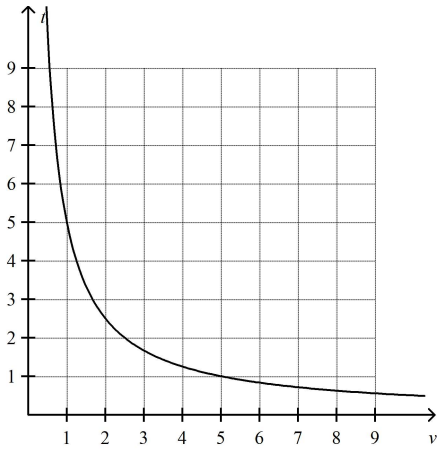
PROBLEM

1. The time, t, in hours, that it takes Alistair to jog 5 km is inversely proportional to his average speed, v, in kilometres per hour.
a) Write a function to represent the time as a function of the speed.
b) Sketch the graph of this function.
c) If Alistair jogs at 4.5 km/h, how long does it take him to complete a 5-km run, to the nearest minute?

ANS:

a) $t(v) = \frac{5}{v}, v > 0$

b)



c) 1 h and 7 min

DIF: Easy OBJ: Section 9.1 | Section 9.3 NAT: RF14
TOP: Exploring Rational Functions Using Transformations | Connecting Graphs and Rational Equations KEY: word problem | reciprocal of linear function | graph

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2. The pressure exerted on the floor by the heel of someone's shoe is inversely proportional to the square of the width of the heel of the shoe. When Megumi wears 2-cm-wide heels, she exerts a pressure of 400 kPa.

a) Determine a function to represent the pressure, p, exerted by Megumi if she wears heels of width w.

b) Sketch the graph of this function.

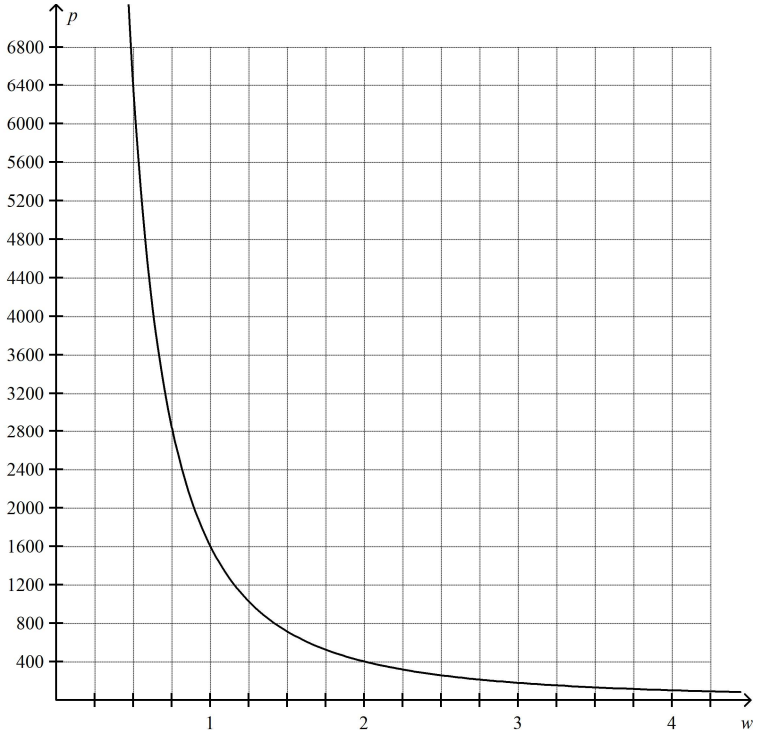
c) If she wears spike heels with a width of 0.5 cm, what pressure does she exert?

ANS:

a) Since the pressure, p, exerted by a person's shoe is inversely proportional to the width, w, of the shoe, you can write $p(w) = \frac{k}{w^2}$ for some k. Since $p(2) = 400$, substituting these values into $p(w) = \frac{k}{w^2}$ gives $k = 1600$.

$p(w) = \frac{1600}{w^2}$, $w > 0$

b)



c) $p(w) = \frac{1600}{w^2}$

$p(0.5) = \frac{1600}{0.5^2}$

$= 6400$

Megumi exerts 6400 kPa of pressure.

DIF: Average OBJ: Section 9.3 NAT: RF14
TOP: Connecting Graphs and Rational Equations
KEY: reciprocal of quadratic function | graph

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3. A photographer uses a light meter to measure the intensity of light from a flash bulb. The intensity, I, in lux, of the flash bulb is a function of the distance, d, in metres, from the light and can be represented by $I(d) = \frac{10}{d^2}$, $d > 0$.

a) Determine the following, to two decimal places:

i) the intensity of light 3 m from the flash bulb

ii) the average rate of change in the intensity of light for the interval $1 < d < 3$

b) What does the sign of your answer to part a)ii) indicate about the light intensity?

ANS:

a) i) $I(d) = \frac{10}{d^2}$

$$I(3) = \frac{10}{3^2}$$

$$\approx 1.11$$

The intensity is 1.11 lux.

ii) $I(d) = \frac{10}{d^2}$

$$\frac{I(3) - I(1)}{3 - 1} = \frac{\frac{10}{3^2} - \frac{10}{1^2}}{2}$$

$$= -4.44$$

The rate of change of intensity is –4.44 lux/m.

b) As the distance from the light increases, the intensity decreases.

DIF: Average OBJ: Section 9.1 | Section 9.3 NAT: RF14
TOP: Exploring Rational Functions Using Transformations | Connecting Graphs and Rational Equations KEY: reciprocal of quadratic function | average rate of change

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4. The population, P, of a small town is modelled by the function $P(t) = \frac{20(4t+3)}{2t+5}$, where t is time, in years, since the start of 1990.

a) Is the population increasing or decreasing? Justify your answer.

b) The town will need its own transit system if the population exceeds 50 000. Will the town's population ever exceed 50 000? Explain.

ANS:

a) There are several ways to determine this result. One way involves approximating the instantaneous rate of change for the current year. For 2012, $t = 22$ and $m \doteq 0.12$. Since $m > 0$, the population must be increasing. Another way is to determine the population at different times, such as at $t = 21$ and $t = 22$, and show that the population is going up. A graph could also be used.

b) There are several ways to determine this result. There is a horizontal asymptote at $P = 40$. Thus, the population never exceeds 40 000. Another way is to show that $\frac{20(4t+3)}{2t+5} > 50$ has no solution for positive t-values. A graph could also be used.

DIF: Difficult + OBJ: Section 9.1 | Section 9.3 NAT: RF14
TOP: Exploring Rational Functions Using Transformations | Connecting Graphs and Rational Equations KEY: rational inequality | increasing | decreasing

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5. Ingrid is docking a motorboat. She turns off the power and lets the boat coast toward the dock. The distance, d, in metres, between the boat and the dock as a function of time, t, in seconds, is given by $d(t) = \frac{-7t + 70}{t + 4}$, $0 \le t \le 10$.

a) What is the average velocity of the boat during the time interval from when Ingrid turns the boat off to when it meets the dock?
b) Determine the velocity of the boat when Ingrid turns off the power, to one decimal place.
c) Determine the velocity of the boat when it meets the dock, to one decimal place.
d) Sketch the graph of the function. On the same set of axes, sketch the graph of the *speed* of the boat in relation to the time.

ANS:
a) Ingrid turns off the boat at $t = 0$ and the boat hits the dock when $d = 0$. Set $d = 0$ and solve for t:

$$d(t) = \frac{-7t + 70}{t + 4}$$

$$0 = \frac{-7t + 70}{t + 4}$$

$$0 = -7t + 70$$

$$0 = -t + 10$$

$$t = 10$$

Thus, $t = 10$ when the boat hits the dock. To find the average velocity, determine $\frac{\Delta d(t)}{\Delta t}$ for $0 \le t \le 10$:

$$\frac{\Delta d(t)}{\Delta t} = \frac{d(10) - d(0)}{10 - 0}$$

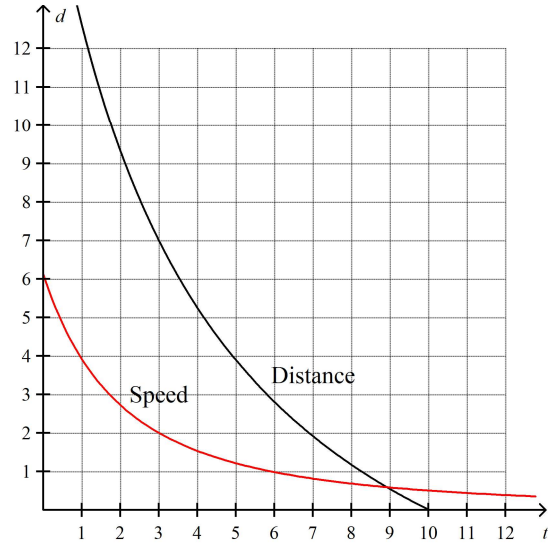
$$= \frac{0 - \frac{70}{4}}{10}$$

$$= -1.75$$

The average velocity is 1.75 m/s toward the dock.
b) 6.1 m/s toward the dock
(This is the instantaneous velocity at $t = 0$.)
c) 0.5 m/s toward the dock
(This is the instantaneous velocity at $t = 10$.)

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d)



The red line is the graph of the *speed* of the boat with respect to its starting position. It is decreasing because the speed of the boat decreases once power has been cut. Students should not be expected to find this graph precisely but should intuit its basic shape.

Note: Although $d'(t) = -\frac{98}{(t+4)^2}$ is negative, this derivative does not represent the speed of the boat, only the rate of change in the distance between the boat and the dock, which is shrinking. In fact, the equation for the speed of the boat with respect to its starting position is the exact opposite of this: $speed = \frac{98}{(t+4)^2}$, which is the equation of the red line graphed above. Again, students are not expected to be this precise.

DIF: Difficult + OBJ: Section 9.1 | Section 9.3 NAT: RF14
TOP: Exploring Rational Functions Using Transformations | Connecting Graphs and Rational Equations KEY: linear expressions in numerator and denominator | rate of change | graph

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6. **a)** Write an equation to represent a rational function with the following conditions:
• reciprocal of a quadratic function

• asymptotes with equations $x = -5$, $x = \frac{4}{3}$, and $y = 0$

• $y < 0$ whenever $x < -5$ or $x > \frac{4}{3}$

b) How many such equations are there? Explain your answer.

ANS:
a) Answers may vary. Sample answer:

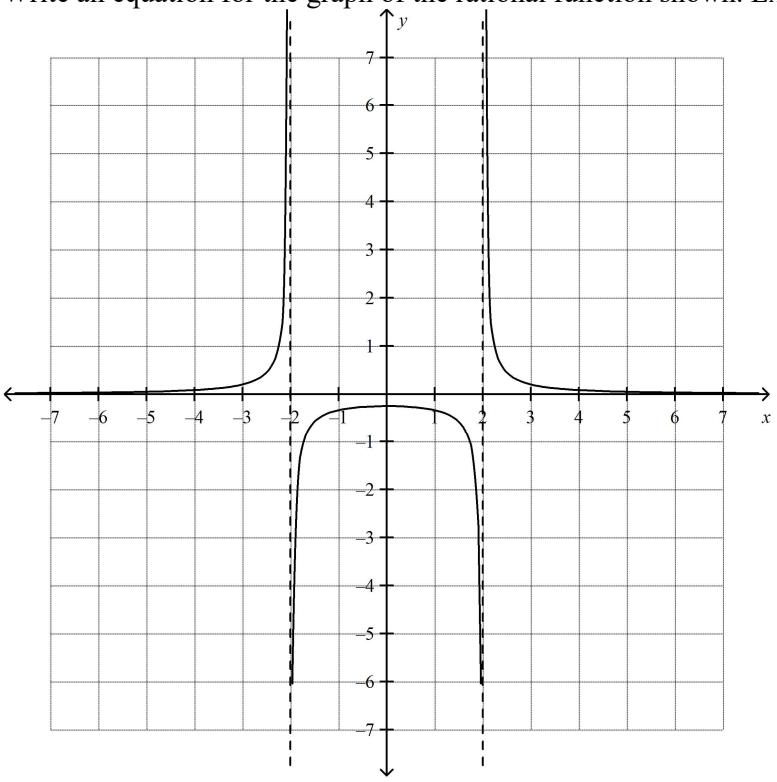
$$f(x) = -\frac{1}{3x^2 + 11x - 20}$$

b) There are infinitely many equations. Any function of the form $f(x) = -\frac{k}{3x^2 + 11x - 20}$, $k > 0$, will satisfy the conditions.

DIF: Difficult OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions KEY: reciprocal of quadratic function

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7. Write an equation for the graph of the rational function shown. Explain your reasoning.



ANS:
Answers may vary. Sample answer:

$$f(x) = \frac{1}{x^2 - 4}$$

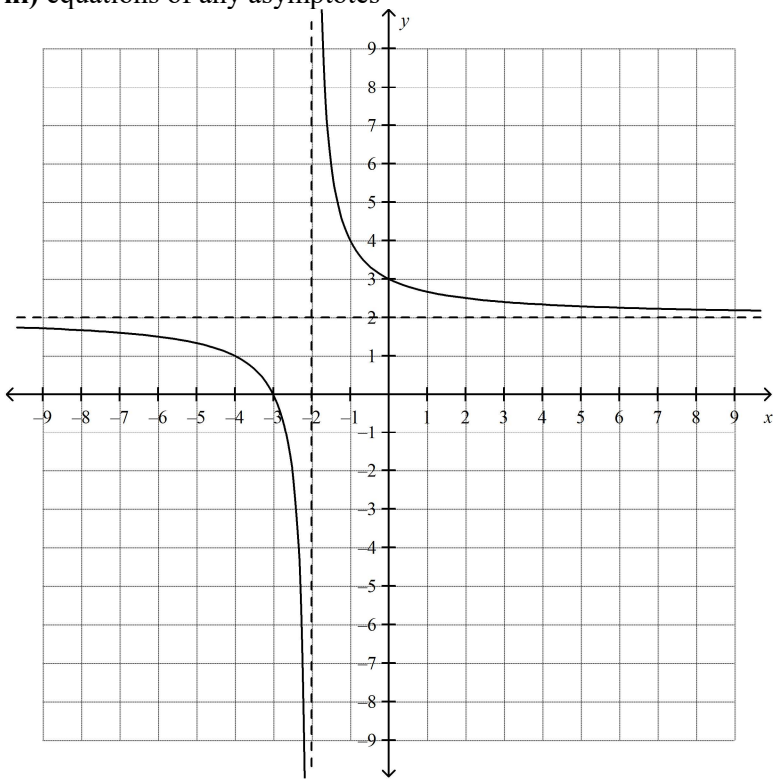
Any function of the form $f(x) = \frac{k}{x^2 - 4}$, $k > 0$, is a reasonable candidate since it is difficult to tell from the graph how stretched the function is.

DIF: Average OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions KEY: reciprocal of quadratic function

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8. **a)** The graph of a function of the form $f(x) = \dfrac{ax+b}{cx+d}$ is shown. Determine the key features:

i) domain and range
ii) intercepts
iii) equations of any asymptotes



b) State a possible equation for this function.

ANS:
a) i) $\{x|x \neq -2, x \in \mathrm{R}\}, \{y|y \neq 2, y \in \mathrm{R}\}$
ii) x-intercept: –3, y-intercept: 3
iii) $x = -2$, $y = 2$
b) Answers may vary. Sample answer:
$f(x) = \dfrac{2x+6}{x+2}$
(Since there is a vertical asymptote at $x = -2$, the denominator must be a multiple of $x + 2$. Since there is a horizontal asymptote at $y = 2$, the ratio $\dfrac{a}{c} = 2$. If you pick $c = 1$, then $a = 2$. To ensure an x-intercept of –3 and a y-intercept of 3, $b = 6$.)

DIF: Average OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions
KEY: linear expressions in numerator and denominator | graph | key features

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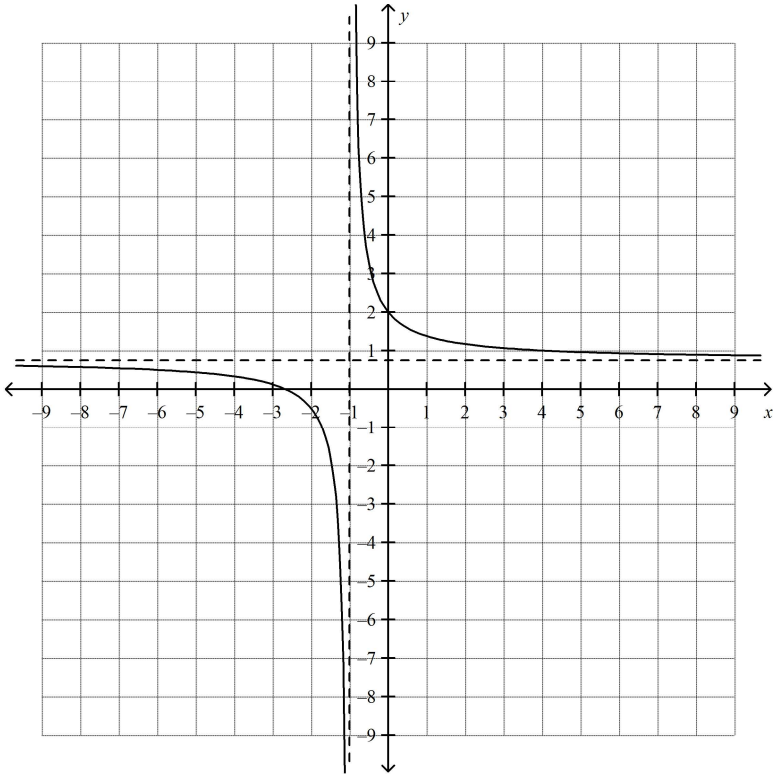
9. **a)** Determine an equation in the form $f(x) = \frac{ax+b}{cx+d}$ for a function that has asymptotes with equations $x = -1$ and $y = \frac{3}{4}$ and a y-intercept of 2.

b) Sketch the graph of your function.

ANS:
Answers may vary. Sample answer:

a) $f(x) = \frac{3x+8}{4x+4}$

b)



(Since there is a vertical asymptote at $x = -1$, the denominator must be a multiple of $x + 1$. Since there is a horizontal asymptote at $y = \frac{3}{4}$, the ratio $\frac{a}{c} = \frac{3}{4}$. If you pick $c = 4$, then $a = 3$. To ensure a y-intercept of 2, $d = 4$ and $b = 8$.)

DIF: Difficult OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions
KEY: linear expressions in numerator and denominator | graph

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10. Explain the difference between the solution to the equation $\frac{4x}{x-2} = 3$ and the solution to the inequality $\frac{4x}{x-2} \le 3$.

ANS:

The equation $\frac{4x}{x-2} = 3$ has one solution, $x = -6$. However, there are infinitely many solutions to $\frac{4x}{x-2} \le 3$, with $x = -6$ being one of them, acting as a boundary on the interval of solutions for x.

DIF: Difficult + OBJ: Section 9.1 | Section 9.3 NAT: RF14
TOP: Exploring Rational Functions Using Transformations | Connecting Graphs and Rational Equations KEY: rational inequality | rational equation

11. **a)** Use long division to express the function $f(x) = \frac{x^2 - 3x + 7}{x + 2}$ as the sum of a polynomial function and a rational function of the form $\frac{A}{x+2}$, where A is a constant.

b) What does your result in part a) allow you to predict about the graph of $f(x) = \frac{x^2 - 3x + 7}{x + 2}$, for large values of x? Why?

ANS:

a) $f(x) = x - 5 + \frac{17}{x+2}$

b) There is a slant asymptote with equation $y = x - 5$. As $x \to \infty$, $\frac{17}{x+2} \to 0$, and so $f(x) = x - 5 + \frac{17}{x+2} \to x - 5$, which is a straight line.

DIF: Difficult + OBJ: Section 9.2 NAT: RF14
TOP: Analysing Rational Functions KEY: slant asymptote | polynomial long division

12. Write an equation for a rational function whose graph has all of the following features:
- vertical asymptote with equation $x = 3$
- horizontal asymptote with equation $y = 2$
- hole at $x = 1$
- no x-intercepts

ANS:

Answers may vary. Sample answer:

$f(x) = \frac{2(x-1)^2}{(x-1)(x-3)}$

DIF: Difficult OBJ: Section 9.2 | Section 9.3 NAT: RF14
TOP: Analysing Rational Functions | Connecting Graphs and Rational Equations
KEY: rational function | hole

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13. **a)** Use the asymptotes and intercepts to make a quick sketch of the function $f(x) = \dfrac{x+1}{x-5}$ and its reciprocal $g(x) = \dfrac{x-5}{x+1}$ on the same set of axes.

b) Describe the symmetry in the graphs in part a).

c) Determine the equation of the mirror line in your graph from part a).

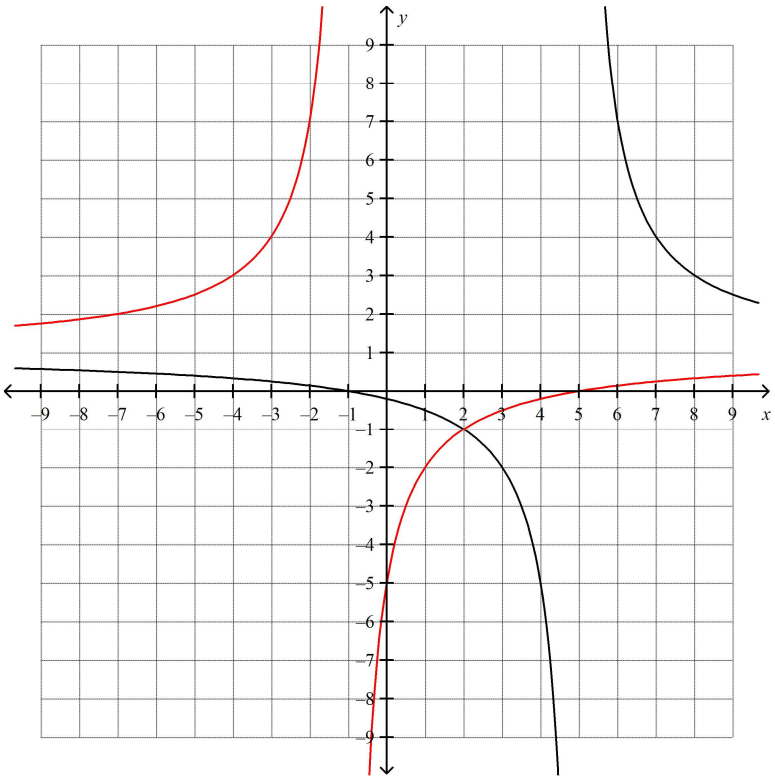
d) Determine intervals where f is positive and where f is negative. Determine intervals where g is positive and where g is negative. How do the two sets of intervals compare to each other?

e) Does the pattern from part d) occur for all pairs of functions $f(x) = \dfrac{x+b}{x+d}$ and $g(x) = \dfrac{x+d}{x+b}$, $b \neq d$? Explain why or why not.

ANS:

a) For $f(x) = \dfrac{x+1}{x-5}$: asymptotes $x = 5, y = 1$; intercepts $(-1,0)$, $\left(0,-\dfrac{1}{5}\right)$

For $g(x) = \dfrac{x-5}{x+1}$: asymptotes $x = -1, y = 1$; intercepts $(5,0)$, $(0,-5)$



b) There is reflective symmetry in this graph, about a vertical mirror line that runs exactly halfway between the two vertical asymptotes.

c) $x = 2$

d) $f(x) > 0$ for $x < -1$ or $x > 5$ and $f(x) < 0$ for $-1 < x < 5$.
$g(x) > 0$ for $x < -1$ or $x > 5$ and $g(x) < 0$ for $-1 < x < 5$.
The intervals where f and g are positive and negative are identical.

e) Yes, this pattern is true for all pairs of functions $f(x) = \dfrac{x+b}{x+d}$ and $g(x) = \dfrac{x+d}{x+b}$ because when you take the reciprocal of any y-value, it does not change sign (i.e., the positive y-values of f remain positive when their reciprocals are taken).

DIF: Difficult OBJ: Section 9.2 | Section 9.3 NAT: RF14

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TOP: Analysing Rational Functions | Connecting Graphs and Rational Equations
KEY: linear expressions in numerator and denominator | graph | reciprocal

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14. **a)** Use the asymptotes and intercepts to make a quick sketch of the function $f(x) = \frac{x-2}{x+4}$ and its reciprocal, $g(x) = \frac{x+4}{x-2}$, on the same set of axes.

b) Describe the symmetry in the graphs in part a).

c) Determine the equation of the mirror line in your graph from part a).

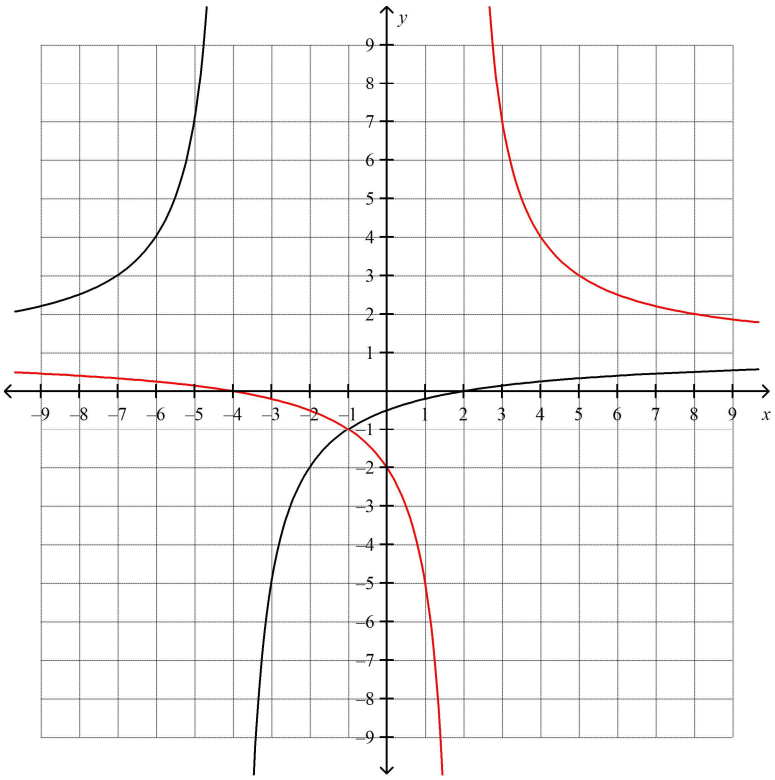
d) Determine intervals of increase and decrease for both f and g. How do the sets of intervals compare?

e) Does the pattern from part d) occur for all pairs of functions $f(x) = \frac{x+b}{x+d}$ and $g(x) = \frac{x+d}{x+b}$, $b \neq d$? Explain why or why not.

ANS:

a) For $f(x) = \frac{x-2}{x+4}$: asymptotes $x = -4, y = 1$; intercepts $(2,0)$, $\left(0, -\frac{1}{2}\right)$

For $g(x) = \frac{x+4}{x-2}$: asymptotes $x = 2, y = 1$; intercepts $(-4,0)$, $(0,-2)$



b) There is reflective symmetry in this graph, about a vertical mirror line that runs exactly halfway between the two vertical asymptotes.

c) $x = -1$

d) For f: increasing for $x \in \mathrm{R}, x \neq -4$, decreasing for no values of x

For g: increasing for no values of x, decreasing for $x \in \mathrm{R}, x \neq 2$.

Where f increases, g decreases, except at $x = -4$ and $x = 2$, the locations of the vertical asymptotes.

e) Yes, this pattern occurs for all such pairs of functions. Values that are growing larger grow smaller when their reciprocals are taken, and vice versa.

DIF: Difficult + OBJ: Section 9.2 | Section 9.3 NAT: RF14
TOP: Analysing Rational Functions | Connecting Graphs and Rational Equations
KEY: linear expressions in numerator and denominator | graph | increasing | decreasing | reciprocal

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15. Solve each inequality by graphing, using technology. Express your answers to one decimal place.

a) $\frac{2x+1}{x-3} \le x$

b) $\frac{x}{x+5} - \frac{1}{x} < 2$

c) $\frac{3x^2-4}{x^2-9} \ge -\frac{1}{x^2}$

d) $\frac{2x}{x-3} > \frac{3x}{2x+4}$

ANS:

a) $-0.2 \le x < 3$ or $x \ge 5.2$

b) $x < -10.5$ or $-5 < x < -0.5$ or $x > 0$

c) $x < -3$ or $-1.5 \le x < 0$ or $0 < x \le 1.5$ or $x > 3$

d) $x < -17$ or $-2 < x < 0$ or $x > 3$

DIF: Difficult + OBJ: Section 9.3 NAT: RF14
TOP: Connecting Graphs and Rational Equations
KEY: rational inequality | technology

16. Solve.

a) $\frac{6}{x+3} > 2$

b) $\frac{2x+4}{x-2} \ge 0$

c) $\frac{x^2-4x-12}{x^2-25} \ge 0$

d) $\frac{x}{x+4} < \frac{x+4}{x}$

ANS:

a) $-3 < x < 0$

b) $x \le -2$ or $x > 2$

c) $x < -5$ or $-2 \le x < 5$ or $x \ge 6$

d) $-4 < x < -2$ or $x > 0$

DIF: Difficult + OBJ: Section 9.3 NAT: RF14
TOP: Connecting Graphs and Rational Equations KEY: rational inequality

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17. Solve.

a) $x+2=\dfrac{2}{x+3}$

b) $\dfrac{2x}{x-2}=2-\dfrac{1}{x+2}$

c) $\dfrac{2x+1}{x+2}=\dfrac{x+3}{x+4}$

d) $\dfrac{1}{x}+\dfrac{1}{x^2}=-\dfrac{1}{x^3}$

ANS:

a) $x = -4$ or $x = -1$

b) $x=-\dfrac{6}{5}$

c) $x=-2\pm\sqrt{6}$

d) No solution.

DIF: Average OBJ: Section 9.3 NAT: RF14
TOP: Connecting Graphs and Rational Equations KEY: rational equation

18. An airplane makes a 990-mi flight with a tail wind and returns, flying into the wind. The total flying time is 3 h 20 min, and the plane's airspeed is 600 mph. What is the wind speed?

ANS:

Let w represent the wind speed.

Then, $600+w$ is the ground speed with the wind

$600-w$ is the ground speed against the wind

3 h 20 min is equal to $\dfrac{10}{3}$ h.

Therefore,

$$\frac{990}{600+w}+\frac{990}{600-w}=\frac{10}{3}$$

$$(3)990(600-w)+(3)990(600+w)=10(600+w)(600-w)$$

$$(3)99(600-w)+(3)99(600+w)=(600+w)(600-w)$$

$$178\,200-297w+178\,200+297w=360\,000-w^2$$

$$w^2=3600$$

$$w=60$$

The wind speed is 60 mph.

DIF: Average OBJ: Section 9.3 NAT: RF14
TOP: Connecting Graphs and Rational Equations KEY: rational equation

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19. What is the solution to the rational equation $\frac{x-2}{x+4}+\frac{x+1}{x+6}=\frac{11x+32}{x^2+10x+24}$?

ANS:

$$\frac{x-2}{x+4}+\frac{x+1}{x+6}=\frac{11x+32}{x^2+10x+24}$$

$$\frac{x-2}{x+4}+\frac{x+1}{x+6}=\frac{11x+32}{(x+4)(x+6)}$$

$$(x+4)(x+6)\left(\frac{x-2}{x+4}+\frac{x+1}{x+6}\right)=\left(\frac{11x+32}{(x+4)(x+6)}\right)(x+4)(x+6)$$

$$(x-2)(x+6)+(x+4)(x+1)=11x+32$$

$$x^2+4x-12+x^2+5x+4=11x+32$$

$$2x^2-2x-40=0$$

$$x^2-x-20=0$$

$$(x-5)(x+4)=0$$

$$x=5,\ x=-4$$

Check:

Let $x=5$.

$$\text{L.S.}=\frac{5-2}{5+4}+\frac{5+1}{5+6}=\frac{3}{9}+\frac{6}{11}=\frac{33}{99}+\frac{54}{99}=\frac{87}{99}$$

$$\text{R.S.}=\frac{11(5)+32}{5^2+10(5)+24}=\frac{87}{99}$$

L.S. = R.S.

The value of x cannot be -4, because $x=-4$ is an inadmissable value.

Therefore, the solution is $x=5$.

DIF: Difficult OBJ: Section 9.3 NAT: RF14
TOP: Connecting Graphs and Rational Equations
KEY: quadratic denominator | extraneous solution

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20. A ski club charters a bus for a ski trip at a cost of $480. In an attempt to lower the bus fare per skier, the club invites non-members to go along. After five non-members join the trip, the fare per skier decreases by $4.80. How many club members are going on the trip?

ANS:
Let x represent the number of members going on the ski trip.

$$\frac{480}{x} - \frac{480}{x+5} = 4.8$$

$$(x)(x+5)\left(\frac{480}{x} - \frac{480}{x+5}\right) = 4.8(x)(x+5)$$

$$480(x+5) - 480x = 4.8(x)(x+5)$$

$$100(x+5) - 100x = x(x+5)$$

$$100x + 500 - 100x = x^2 + 5x$$

$$x^2 + 5x - 500 = 0$$

$$(x+25)(x-20) = 0$$

$$x = -25,\ x = 20$$

The negative answer cannot be used in the context of this problem, and it is excluded.
20 members of the ski club are going on the trip.

DIF: Average OBJ: Section 9.3 NAT: RF14
TOP: Connecting Graphs and Rational Equations
KEY: linear expressions in numerator and denominator | x-intercept | excluded solution

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